

How Can We Design Playful Technology Aimed at Promoting Audience Participation in Museums?

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Abstract

The present times are witnessing a transformation of museums into more participatory spaces where visitors can create, share, and connect with each other around content. Excited about what this shift might mean for HCI research, I approach it from a perspective of play and playfulness, exploring how to design playful technology to promote audience participation in museums. I adopt Research through Design as the core methodology, aiming to generate intermediate-level knowledge that foregrounds the potential of such technology in museums and provides actionable guidance for realizing that potential by design. In this paper, I present an overview of my project, including: outcomes of the first case study I have completed so far, ongoing efforts to analyze an annotated portfolio of existing museum installations, and my plans toward completing the Ph.D. I look forward to discussing my work with peers and mentors to enrich my thoughts on approaching my upcoming work.

CCS Concepts

• Human-centered computing; • Interaction design;

Keywords

Playful technology, Museum experiences, Audience participation, Research through design, Co-design

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1 Introduction and Background

Museums, as multicultural living spaces, are evolving from contemplative to participatory spaces. In turn, audience engagement also shifts from a paradigm of “receptive” involvement to a more “participatory” one [14] – from unidirectional spectating to arenas to “where visitors can create, share, and connect with each other around content” [13]. Nina Simon, a recognized scholar and museum curator, describes participatory museums as “meeting grounds for dialogue” [13], in which institutions actively collect and share

diverse, personalized, and evolving content co-generated with visitors, while also inviting visitors to respond to and discuss that content. Building on this premise, my doctoral research looks at how interactive technologies can play a role in supporting such “meeting grounds for dialogue.”

In HCI, the design and implementation of interactive technology to increase audience engagement in museums is far from novel [5]; researchers have long explored it with diverse, sometimes compatible agendas, such as delivering customized experiences [9], facilitating social connection [7], etc. Among this body of work, I noticed particularly exciting studies that explore how interactive technologies can facilitate dialogue, debate, and participation among multiple stakeholders within museum experiences. An example is Kambunga et al.’s experimental participatory exhibition [6], which explores how interactive installations can foster meaningful dialogue between museum audiences and youth, generating new perspectives on postcolonialism. In all, the shift toward increasingly participatory museums opens up new opportunities for HCI research.

Through reviewing related literature from HCI and beyond, I am attentive to the fact that the existing studies rarely examine the underlying qualities of play and playfulness entangled with participatory audience engagements. That is particularly true in relation to play’s *eudaimonic* capacity [11], i.e., its potential to be socially, emotionally, and culturally productive, even if it does not lead to tangible outcomes. In particular, these ambitions closely align with those of participatory museums, insofar as they are seen as spaces for community building, for bonding, for individual and collective growth [13]. In response to this gap, I approach the space of museum participation in HCI from a specific angle: exploring how such a participatory move in museums could be supported through play and playfulness.

Building on that, I articulated the overarching research aim and question for my doctoral study: *how can we design playful technology aimed at promoting audience participation in museums?* To answer it, I conducted a case study with *Museu del Cinema*, where I sought to understand which forms of playful interaction and participation different stakeholders of that particular museum longed for, and began to speculate on how to respond to them by design. Then, I consolidated the contribution of the first case study and reflected on its limitations, re-anchoring and narrowing the research focus toward the social, dialogical, and communitary layers of audience participation in museums. Currently, I am working on an *annotated portfolio* [8] to begin to characterize (and make actionable) this novel design space within museum-related HCI that I call “playgrounds for dialogue.” In future work, I will build



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on the annotated portfolio findings in the second case study, where I will engage another museum and its stakeholders to co-design a “playground for dialogue” as a means to consolidate this novel genre as a *strong concept* [4]. The culmination of my doctoral work will be the formulation of a design framework for “playgrounds for dialogue,” to guide designers in creating technologies that enable collective audience participation in museums by playful means.

At this stage, I look forward to engaging in discussions with peers and mentors at CHI about my work, consolidating my understanding of the research I have completed, and enriching my thinking about my upcoming efforts.

2 Methodology and Overview of Doctoral Research

In my doctoral project, I examine how we might design playful technology that contributes to empowering audiences to more actively, meaningfully, and delightfully participate in museum experiences. To do this, I situate myself in the context of real museums within my local community and co-imagine with diverse stakeholders what kinds of playful technologies might give rise to the forms of audience participation these museums and communities long for. Overall speaking, my work adopts Research through Design [3] as the core methodology, aiming to generate intermediate-level [8] knowledge that speaks to the potential of playful technology to support audience participation in museums and provides actionable inspiration for realizing that potential by design. Through a combination of case studies of my own work and analyses of other designers’ relevant productions, I seek to derive resulting ideas that hold generative value beyond a particular design case: insights that provide actionable values for design research that assist designers in considering when developing playful technologies aimed at enhancing audience participation in museums.

Here I summarize the broad structure of my doctoral project, whose different phases I unpack in more depth in Sections 3–5. Taking my research question as a point of departure, I began to approach this design space through a case study with *Museu del Cinema* (to be elaborated in Section 3). Starting in April 2024, I conducted a one-year-long Participatory [10] Research through Design [3] project, engaging diverse stakeholders of the museum in an open-ended, bottom-up way to co-image playful technologies that would invite people to engage more actively within the museum. From a reflexive analysis of the design process and its results, I derived a two-fold contribution: (1) a catalog featuring three design ideas showcasing different ways of active audience participation; (2) a set of design insights for creating such technologies.

Through the lens of literature on participatory museums, I retrospectively reflected upon the contribution of the first case study, and found that it covered only a fraction of what participation might mean in the context of museums – though the entire process allowed the voices of the participants I engaged with to determine the outcomes of the exploration. Specifically, the outcomes remained at the individual engagement level rather than foregrounding social connections among visitors to transform it into a collective one. That led me to decide to further explore playful technology on the social, dialogical, and communitary layers of audience participation.

In my current study (from November 2025), I began calling this exciting design space “playgrounds for dialogue,” to illustrate what it might look like, identify diverse ways it might manifest, and start to make it actionable. In Section 4, I present an in-progress analysis of an *annotated portfolio* [8] of museum installations that support collective activity through play and playfulness. From this, I am beginning to surface a set of social affordances that might sustain “playgrounds for dialogue” and starting to unearth design considerations designers might want to attend to when creating such technologies.

The outcomes of this in-progress work will support my efforts to promote my second case study (to be initiated by Spring 2026). Through situating myself in the context of a second museum and co-designing with its stakeholders, I will consolidate “playgrounds for dialogue” as a *strong concept* [4] and lead it to the formulation of a design framework that offers actionable knowledge for designers when creating technologies that enable audience active, collective participation. I plan to start writing my final dissertation in October 2026 and complete my doctoral project by July 2027. In the following sections, I elaborate on these phases of my work in detail, covering both my completed, current, and future doctoral work.

3 Completed Work: Exploratory Case Study with *Museu del Cinema*

As a point of departure, I began to approach the design space of playful technology for participatory museums through a first case study. In this work, my ambitions were deliberately open-ended: I adopted a broad definition of what “playful technology” and “participation” might mean, allowing the voices of the participants I engaged with to determine the outcomes of the exploration. Through that open-ended, bottom-up process, I sought to understand what forms of playful interaction and participation different stakeholders of a particular museum longed for and began to speculate how to respond to them by design.

To anchor that exploration in a specific setting, this project collaborated with *Museu del Cinema* (the official website: <https://museudelcinema.girona.cat/>). It is a local cultural institution dedicated to conserving, researching, and interpreting the Tomàs Mallol Collection and other film-related collections or objects, with the goal of increasing public understanding of the history of moving images and cinema. Within that context, I facilitated a multi-phased participatory process where I promoted dialogue between different stakeholders of *Museu del Cinema* to co-design playful technologies aimed at promoting audience participation in experiences. Figure 1 presents a visual synthesis of the research process.

The project began with a phase of **contextual research (Phase 1)** aimed at familiarizing me with *Museu del Cinema* and gathering insights to be used as foundations for subsequent co-design efforts. To do that, I learned about *Museu del Cinema* as a space and all the content showcased in the permanent exhibition. As well, I undertook two further actions, oriented toward data collection: a structured questionnaire for museum visitors and an interview with museum staff. The contextual research led me to derive early insights into what audiences of *Museu del Cinema* and museum staff long for in technology-mediated experiences.

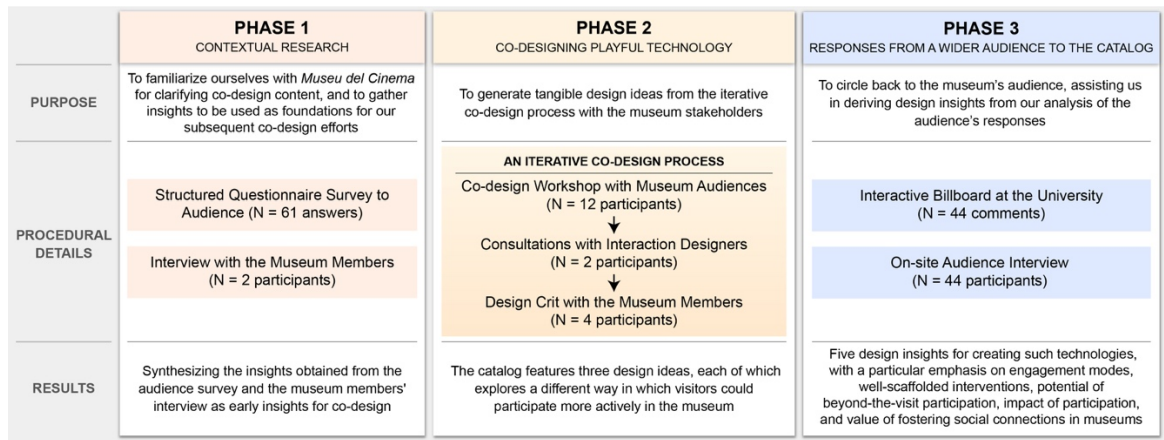


Figure 1: The research flowchart outlines the purposes, procedural details, and results for each phase.

Based on those early insights, I initiated the second phase, **co-designing playful technology (Phase 2)**, which took the form of an iterative co-design process aimed at generating tangible design ideas that could support audience participation in playful ways. To do so, I invited diverse stakeholders to co-design with us: first, a co-design workshop with museum audiences; then, consultations with other interaction designers; and finally, a design crit involving museum members.

The iterative process carries the expertise and viewpoints of these different stakeholders, from which we generate three design ideas for *Museu del Cinema*. Here I describe them as an *annotated portfolio* [8], which can be downloaded at: <https://bit.ly/4pVFmQB>. Of which illustrates different ways museum audiences could participate more actively: Idea 1, *My Images Start Moving*, allows visitors to generate new content in two ways of encouraging creation and displaying their likenesses to become part of the exhibition; Idea 2, *Silent Film Stars*, guides audiences to participate in a way of personalized quiz, then display their photos as engagement outcomes in the museum; and Idea 3, *To Find Fun*, features clear rules to engage visitors to participate through game-like mechanics. These design ideas in the portfolio can inspire cultural institutions and designers alike to implement similar technologies or activities, hinting at promising directions for blending play with participation in museums.

To **get responses to the catalog from a wider audience (Phase 3)**, I set up an interactive billboard at my university that invited passersby who had visited the museum to share their opinions on the three ideas, and conducted on-site interviews at the museum where I invited visitors to browse the catalog of ideas and engaged them in conversations about their reactions. Afterwards, I reflexively [12] analyzed data from both sources – the interactive billboard and the on-site interviews – together, identifying these five design insights I could pass along to others interested in this design space:

- *Multiplicitous entry points*: Technology should support multiple ways of playing to accommodate the idiosyncrasies of diverse museum audiences.
- *Lightweight scaffolds for participation*: Well-scaffolded interventions could assist the delivery of lightweight experiences,

involving in a relaxed and joyful way in being involved in the experience.

- *Beyond-the-visit carryover by playful technology*: Playful technology can promote on-site audience participation, as well as inspire audiences to actively extend their participation beyond their visit.
- *Making durable impacts by ephemeral interactions*: Audiences look forward to more permanent participation impacts through transitory interactions with playful technology.
- *Social connection by design*: Playful technologies are highly valuable for fostering social connection in museums, which can help museums develop concrete aims, such as fostering dialogue, building relationships, and encouraging sharing.

These insights can guide the design of playful installations that enhance audience participation in museums, assisting designers in their decision-making in a real implementation case. Overall, I present a two-fold contribution from a reflexive analysis of our design process and the results: (1) a catalog featuring three design ideas of playful technologies for *Museu del Cinema*, which show-cases different ways in which technology might support museum audiences to participate more actively; and (2) a series of insights derived from the multi-stakeholder conversations around the catalog, which might inspire designers interested in creating playful technologies that supporting audience participation in museums. Though these contributions stemmed from a case study of *Museu del Cinema*, anchored in a specific socio-cultural context and involving a specific set of stakeholders, I believe the resulting ideas hold generative value beyond this case.

In retrospectively reflecting upon the contributions, I realize that while this work explored ways of approaching the consumption of existing content from an interactive (rather than passive) stance, these are but a fraction of what participation might mean in the context of museums. As Simon notes [13], an important ambition of participatory museums is supporting “*individuals [to] engage with each other socially*,” foregrounding social connections among visitors. This is something my first case study did not tackle, at least not explicitly – these social, collective forms of participatory engagement were not reflected in the final outcomes.

I am thus inspired to envision exciting avenues for subsequent work – I expect to explore playful technology-mediated installations that foster social connections among people and collective engagement in museum-related experiences, and begin to call this emerging design space “playgrounds for dialogue,” building on Simon’s vision of participatory museums as “meeting grounds for dialogue” [13]. It clarifies the immediate next step of my Ph.D., leading me to explore playful technology on the social, dialogical, and communitary layers of audience participation.

4 Current Work: Annotated Portfolio of “Playgrounds for Dialogue”

Prior to embarking on case study 2, I started to illustrate what this novel “playgrounds for dialogue” design space might look like, elucidate its bounds, and identify diverse ways it might manifest, aiming to develop it as a *strong concept* [4] – a form of generative, intermediate-level knowledge that resides on an abstraction level above particular instances.

To do that, I am in the process of analyzing existing museum installations, in the form of an *annotated portfolio* [8] of design exemplars, looking closely at what these exemplars do and how they draw people into playful collective behavior. This ongoing work led to a shortlist of 24 design exemplars in an annotated portfolio so far (available for download: <https://bit.ly/477aKF4>). I use a combination of close reading [2][12] and reflexive thematic analysis [1] – examining each installation’s affordances, how each one worked, and available evidence about the experiences it supported, and drawing on my reflexive interpretation to identify themes with potential value for interaction design.

As early in-progress findings from a reflective analysis of the portfolio of exemplars, so far I surfaced three social affordances that can support playful forms of social connections among diverse visitors around museum content: The first social affordance is *People create individual content that contributes to a collective piece*, which means empowering people to create personalized content, rather than simply consuming content curated by museum institutions, and to append it to a shared canvas where others can see, respond to, or sometimes co-create; the second is that *People act individually, yet in co-presence to shape a shared experience*, in which people act individually, without explicit collaboration, yet through co-presence they collectively shape a shared experience; the third affordance is *People gather around to collaborate and act together*, in which people intentionally come together to collaborate and co-act around museum content or activities. Together, these social affordances begin to hint at starting points for designing “playgrounds for dialogue”: they point toward different ways in which social, communitary, dialogical activity can be supported through playful tech.

As well, by looking at granular annotation of the portfolio, I have begun to unearth important design considerations for each social affordance that might assist designers in developing interactive technologies that present these affordances. For example, aiming at Social Affordance #1 – *People create individual content that contributes to a collective piece*, I noticed that facilitating social exchange depends less on what content people create alone than on in which ways individual content is shared with others

and influences others, i.e., how individual creation is bridged to collective contribution. It motivates me to unearth an important consideration on “temporality of participation,” both between individual creation and display and between different participants’ involvement – *How to shorten, or even eliminate, time lags between an individual creating and displaying a contribution to a shared piece? How to support synchronous creation among multiple users?* This part is particularly at an early stage, ongoing, and far from consolidated insights. I plan to deepen and consolidate it by advancing toward case study 2, leading to the formulation of a design framework to make “playgrounds for dialogue” actionable.

5 Future Work: From Case Study 2 to a Design Framework

Based on these in-progress findings on “playgrounds for dialogue,” I am seeking cooperation with a second local museum institution that is interested in fostering social connections among visitors as a field context for my second case study. I will conduct a design research project in the context of that museum, where I plan to design tech ideas for “playgrounds for dialogue” with its stakeholders. Through analysis of this case study, I will generate actionable knowledge for designers when creating such technologies that support social, collective audience participation. Though this second case study is still in an early resourcing phase (i.e., in planning), my rough plan is to initiate the participatory design work by Spring 2026, extend it for a period of 6-8 months leading to a functional prototype, and conduct a naturalistic user study to examine the impact of the resulting design work.

Once the second case study is completed, I integrate my learnings from the three phases of my doctoral work (case studies and annotated portfolio) toward consolidating “playgrounds for dialogue” as a strong concept, and develop a design framework that aims to make it actionable to materialize “playgrounds for dialogue” through playful interaction design. In parallel, I will begin writing my Ph.D. dissertation in October 2026, with the goal of completing the doctoral project by July 2027.

6 Conclusion

In my doctoral research, I use Research through Design to explore how to design playful technology that contributes to empowering audiences to more actively, meaningfully, and delightfully participate in museum experiences. By carrying out two case studies, I aim to generatively contribute design-oriented knowledge that can inform the creation of technology-mediated playful installations promoting audience participation in museums. Drawing on my progress and existing contributions, I am starting to surface “playgrounds for dialogue” as a novel design space within museum-related HCI, and believe that it can be a genre of museum-related interactive design that looks into how to materialize Simon’s ideal of museums as “meeting grounds for dialogue” through playful interaction design.

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